

Week 4: Human Skeleton Detection using MediaPipe

1. Introduction

The objective of this week's assignment was to explore the capabilities of the **MediaPipe** library for real-time skeleton and landmark detection. We implemented four Python scripts to test different models: basic camera access, hand tracking, face mesh detection, and holistic full-body tracking.

2. Exercises Implementation

2.1. Exercise 1: Working with Camera

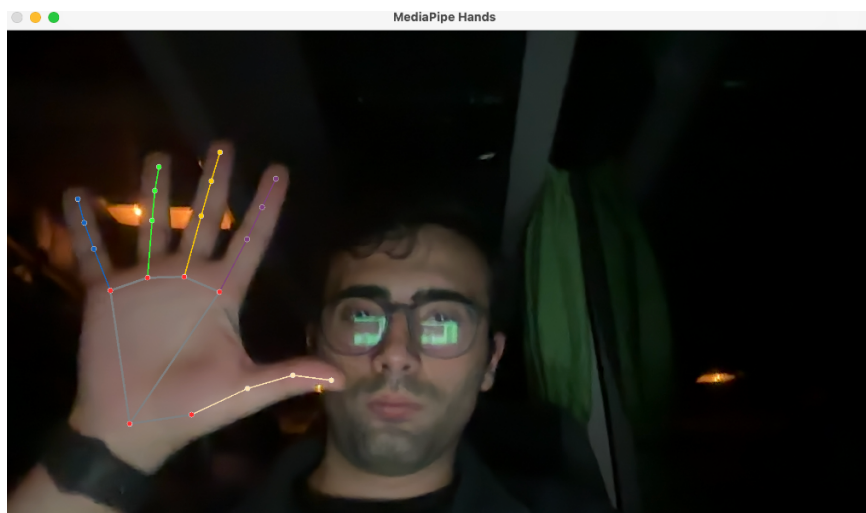
We established a basic computer vision pipeline using OpenCV. The script opens the default camera (ID 0), captures frames in a loop, flips them horizontally to create a mirror effect, and displays them. This serves as the foundation for the subsequent exercises.

2.2. Exercise 2: Hand Detection

We utilized the `mp.solutions.hands` module.

- **Method:** The MediaPipe Hands model processes the RGB frame and returns multi-hand landmarks.
- **Visualization:** We used `mp_drawing.draw_landmarks` to visualize the 21 hand landmarks and the connections between them (skeleton).
- **Result:** The model robustly detects hands even in varying lighting conditions.

Figure 1: Real-time hand landmark detection.



2.3. Exercise 3: Face Mesh

We implemented the `mp.solutions.face_mesh` model.

- **Method:** This model estimates 468 3D face landmarks. We enabled `refine_landmarks=True` to get detailed attention to the eyes and lips.
- **Visualization:** Different styles were applied to contours, tessellations, and irises to distinguish facial features clearly.
- **Result:** The mesh provides a high-density map of the face, suitable for expression analysis.

Figure 2: 468-point Face Mesh detection displaying facial geometry.



2.4. Exercise 4: Holistic Model

We used the `mp.solutions.holistic` model, which unifies separate models for pose, face, and hands.

- **Method:** This comprehensive model yields multiple outputs: pose landmarks (body skeleton), face landmarks, and left/right hand landmarks.
- **Strengths:** It provides a complete understanding of human body language and posture in a single API call.
- **Weaknesses:** It is computationally more expensive than individual models. There can be a slight lag if the hardware is not powerful enough, although on modern

M-series chips it runs smoothly. Accuracy for hands can sometimes decrease if they are far from the body or moving very fast compared to the dedicated hands model.

Figure 3: Holistic detection showing simultaneous tracking of face, hands, and body pose.

